No. of Printed Pages : 4 220731
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3rd Sem / Civil

Subject : Concrete Technology

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 What is the primary purpose of adding air-entraining agents to concrete? (CO1)

a) To increase strength

b) To improve workability

c) To reduce shrinkage

d) To enhance durability

Q.2 Which of the following types of cement is suitable for high-temperature applications? (CO2)

a) Ordinary Portland Cement

b) Rapid Hardening Cement

c) High-Alumina Cement

d) Sulphate-Resisting Cement

Q.3 What is the recommended water-cement ratio for a concrete mix with a slump of 75-100 mm? (CO3)

a) 0.4-0.5 b) 0.5-0.6

c) 0.6-0.7 d) 0.7-0.8

Q.4 What is the purpose of using fly ash in concrete? (CO4)

a) To increase strength
b) To improve workability
c) To reduce shrinkage
d) To enhance durability

Q.5 Which of the following is a type of fiber reinforcement used in concrete? (CO4)

a) Steel fibers
b) Glass fibers
c) Polypropylene fibbers
d) All of the above

Q.6 What is the purpose of curing concrete? (CO5)

a) To increase strength
b) To improve workability
c) To reduce shrinkage
d) To enhance durability

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 The process of adding water to cement to form a paste is called_______ (CO1)

Q.8 The fineness modulus of a sand sample is_______. (CO2)

Q.9 The purpose of curing concrete is to_______. (CO3)

Q.10 The recommended slump for concrete to be placed in a column is_______. (CO4)

Q.11 The type of cement suitable for underwater construction is______. (CO5)

Q.12 The purpose of using a super plasticizer in concrete is to_______. (CO5)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain the types of aggregates used in concrete. (CO1)

Q.14 Explain the concept of water-cement ratio. (CO2)

Q.15 Explain the concept of workability in concrete. (CO2)

Q.16 What are the precautions to be taken during finishing of concrete? (CO3)

Q.17 Describe the differences between hand mixing and machine mixing. (CO3)

Q.18 What are the precautions to be taken during placement of concrete? (CO4)

Q.19 Explain the purpose of curing concrete. (CO4)

Q.20 Explain the types of cement used in high-temperature applications. (CO4)

Q.21 Explain the purpose of using fly ash in concrete. (CO5)

Q.22 Explain the concept of fineness modulus. (CO2)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the principles of concrete mix design. (CO4)

Q.24 Describe the various methods of curing for concrete. (CO5)

Q.25 Write the name of non-destructive tests and explain any one in detail. (CO6)